

1 Write a Python program demonstrating loops with break and continue.

Aim

To write a Python program that demonstrates the use of (to exit a loop early) and (to skip an iteration) in loops.

Algorithm

1. Use a loop to iterate through numbers 1 to 10.
2. Demonstrate :
 - Skip printing even numbers.
3. Demonstrate :
 - Stop the loop completely when the number reaches 7.
4. Print results to show how and affect loop execution.

CODE

Program to demonstrate break and continue in loops

```
print("Demonstrating continue (skip even numbers):")
```

```
for i in range(1, 11):
```

```
    if i % 2 == 0: # Skip even numbers
```

```
        continue
```

```
    print(i, end=" ")
```

```
print("\n\nDemonstrating break (stop at 7):")
```

```
for i in range(1, 11):
```

```
    if i == 7: # Stop loop when i = 7
```

```
        break
```

```
    print(i, end=" ")
```

OUTPUT

Demonstrating continue (skip even numbers):

1 3 5 7 9

Demonstrating break (stop at 7):

1 2 3 4 5 6

Conclusion

- The `continue` statement skips the current iteration and moves to the next one.
- The `break` statement terminates the loop entirely when the condition is met.
- Together, they provide fine control over loop execution in Python

2 Write a program to count vowels in a string.

Aim

To write a Python program that counts the number of vowels (a, e, i, o, u) in a given string.

Algorithm

1. Accept a string from the user.
2. Define a set of vowels (both uppercase and lowercase).
3. Initialize a counter to 0.
4. Traverse each character in the string:
 - If the character is a vowel, increment the counter.
5. Display the total count of vowels.

CODE

Program to count vowels in a string

```
def count_vowels(text):  
    vowels = "aeiouAEIOU"
```

```
count = 0
for char in text:
    if char in vowels:
        count += 1
return count

# Input from user
string = input("Enter a string: ")

# Function call
vowel_count = count_vowels(string)

print(f"Number of vowels in the string: {vowel_count}")
```

OUTPUT

Enter a string: Yogin is learning Python

Number of vowels in the string: 7

Conclusion

- The program successfully counts vowels in a string using a simple loop and condition.
- It works for both uppercase and lowercase vowels.
- This can be extended to also count consonants, digits, or special characters for more detailed analysis.